

Proximal Policy Optimization (PPO)

1 Policy Optimization Basics

1.1 Policy Gradient Methods

The standard policy gradient estimator is defined as:

$$\hat{g} = \hat{\mathbb{E}}_t \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \hat{A}_t \right] \quad (1)$$

where $\pi_{\theta}(a_t | s_t)$ is a stochastic policy parameterized by θ , $\hat{A}_t$ is an advantage estimator at timestep t , and $\hat{\mathbb{E}}_t[\cdot]$ denotes the empirical average over a batch of samples.

The corresponding objective function is:

$$L^{PG}(\theta) = \hat{\mathbb{E}}_t \left[\log \pi_{\theta}(a_t | s_t) \hat{A}_t \right] \quad (2)$$

1.2 Trust Region Policy Optimization (TRPO)

To avoid destructively large steps, TRPO maximizes a surrogate objective subject to a constraint on the average KL divergence:

$$\max_{\theta} \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] \quad (3)$$

$$\text{subject to } \hat{\mathbb{E}}_t [\text{KL} [\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)]] \leq \delta \quad (4)$$

This problem is typically solved using the conjugate gradient method followed by a line search.

It can also be written in unconstrained form using a penalty coefficient β :

$$\max_{\theta} \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t - \beta \text{KL} [\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)] \right] \quad (5)$$

However, choosing a fixed β that works consistently across different tasks is difficult.

2 Clipped Surrogate Objective

Define the probability ratio:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad r_t(\theta_{\text{old}}) = 1 \quad (6)$$

The conservative policy iteration objective is:

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t \right] = \hat{\mathbb{E}}_t [r_t(\theta) \hat{A}_t] \quad (7)$$

To modify the objective, PPO clips the ratio $r_t(\theta)$ to the interval $[1 - \epsilon, 1 + \epsilon]$:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (8)$$

where ϵ is a hyperparameter.

- When $\hat{A}_t > 0$: The objective increases with $r_t(\theta)$ until $r_t(\theta) = 1 + \epsilon$, after which it is clipped.
- When $\hat{A}_t < 0$: The objective decreases with $r_t(\theta)$ until $r_t(\theta) = 1 - \epsilon$, after which it is clipped.

Taking the minimum ensures that we form a lower bound on the unclipped objective.

3 Adaptive KL Penalty

Alternatively, PPO can use an adaptive penalty coefficient β :

$$L^{KLPE}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t - \beta \text{KL} [\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_\theta(\cdot | s_t)] \right] \quad (9)$$

Let d be the mean KL divergence:

$$d = \hat{\mathbb{E}}_t [\text{KL} [\pi_{\theta_{\text{old}}}(\cdot | s_t), \pi_\theta(\cdot | s_t)]] \quad (10)$$

The update rule for β is:

$$\begin{cases} d < d_{\text{targ}}/1.5 \implies \beta \leftarrow \beta/2 \\ d > d_{\text{targ}} \times 1.5 \implies \beta \leftarrow 2\beta \end{cases} \quad (11)$$

4 Combined Loss Function

When using shared architectures between the policy and the value function, the total objective is:

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t [L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t)] \quad (12)$$

where:

- $L_t^{VF}(\theta) = (V_\theta(s_t) - V_t^{\text{target}})^2$ is the squared value function error.
- $S[\pi_\theta](s_t)$ is the entropy bonus used to ensure exploration.
- c_1, c_2 are hyperparameter coefficients.

5 Truncated Advantage Estimation

Running the policy for T timesteps, the truncated advantage estimator is:

$$\hat{A}_t = -V(s_t) + r_t + \gamma r_{t+1} + \dots + \gamma^{T-t-1} r_{T-1} + \gamma^{T-t} V(s_T) \quad (13)$$

where the intermediate terms represent empirical rewards, and $V(s_T)$ is the value prediction at the truncation horizon, discounted by $\gamma \in [0, 1)$.